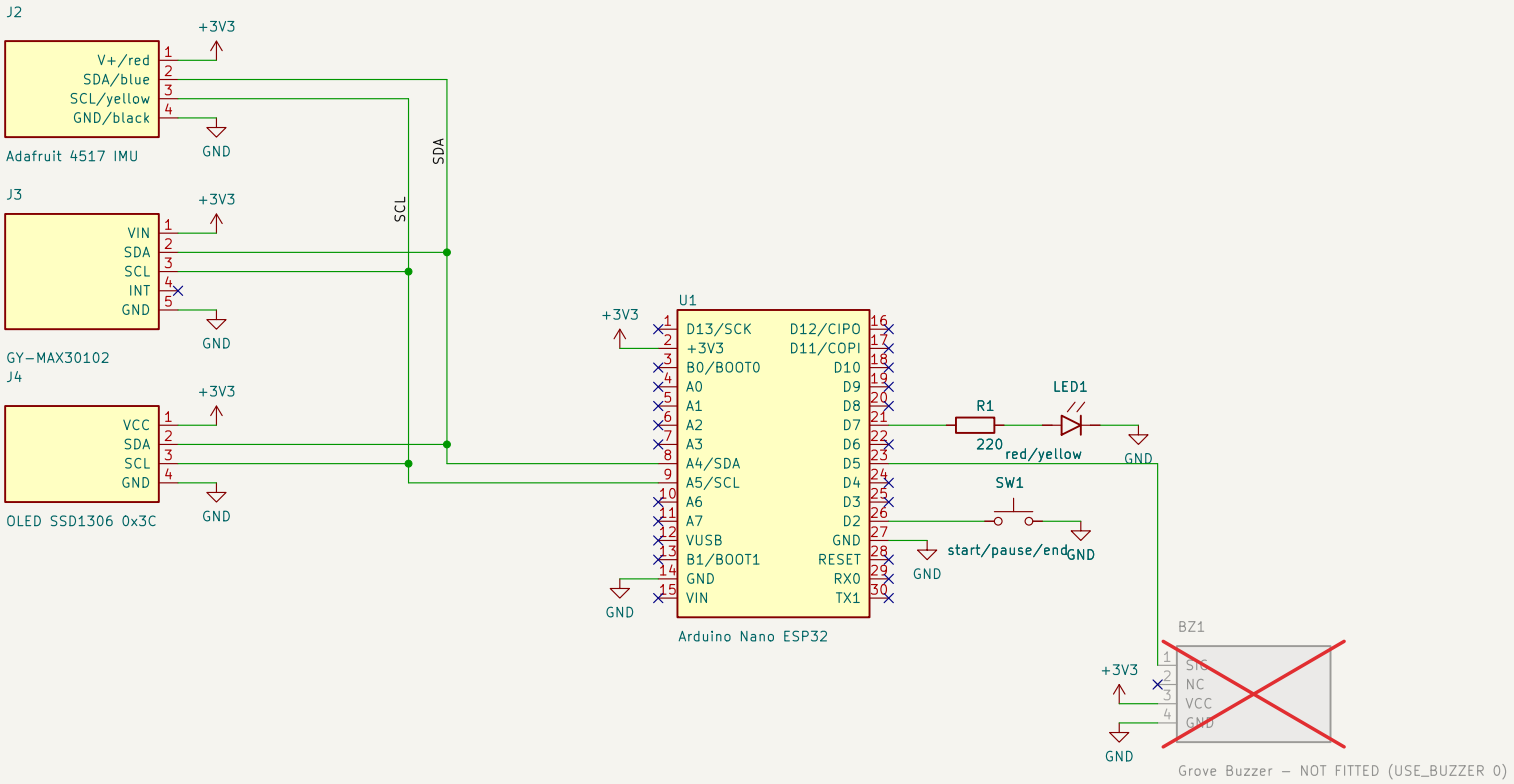



Notes
1. Power: Nano ESP32 only from USB-C (INIU SnapGo P81 power bank). The Nano 3V3 pin feeds every module. VIN and VUSB unused.
2. One I2C bus, 400 kHz: LSM6DSOX 0x6A, LIS3MDL 0x1C (unused), MAX30102 0x57, SSD1306 0x3C. Scanned together 2026-11-03; all four ran in firmware/app 2026-11-06.
3. Pull-ups: 4517 has 10k to 3.3 V; GY-MAX30102 has 4.7k to its own 1.8 V rail. Measure SDA/SCL high level; below 2.5 V remove the MAX30102 pull-ups.
4. SW1: one kit pushbutton to GND, INPUT_PULLUP on D2. Short press start/pause/resume, hold 1.5 s end (firmware/app). Tested 2026-11-06.
5. LED1 on D7 through 220 ohm; use a red or yellow LED at 3.3 V. BZ1 is optional: firmware/app compiles with USE_BUZZER 0, D5 held LOW.
6. 4517 pins follow the 4209 STEMMA QT cable colours. MAX30102 INT and 4517 INT not wired (polling). D8/D9 reserved for them.
Source: hardware/wiring/README.md, firmware/app/app.ino. Redrawn in KiCad from Julia's EasyEDA V2.0 (2026-11-08); regenerate with gen_kicad_sch.py.

Single button, buzzer not fitted. Bench build on half breadboard, Nano off-board.
Drawn: Julia Zhang (EasyEDA V2.0, 2026-11-08). KiCad version: Hanson He.
**ECE 198 Group 57**

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File: smart-dumbbell.kicad_sch

***Title: Smart Dumbbell – wiring v2.1***

| Size: A4 | Date: 2026-11-08 | **Rev: V2.1** |
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| KiCad E.D.A. 10.0.6 | | Id: 1/1 |